Iris Dataset and Artificial Neural Network (ANN)

Beginner Documentation for Kaggle Notebook

Prepared from the uploaded deep-learning.ipynb notebook and the Iris dataset workflow.

1. Introduction

The Iris dataset is a commonly used beginner dataset in machine learning. It contains measurements of iris flowers and their species labels. It is useful for learning data exploration, classification, and model evaluation.

An Artificial Neural Network (ANN) is a machine-learning model inspired by the way biological neurons process information. For this task, an ANN can learn patterns in flower measurements and predict the flower species.

2. Dataset Overview

The Iris dataset contains 150 samples, four numeric input features, and one target label. It has three flower classes: Iris setosa, Iris versicolor, and Iris virginica. Each class has 50 samples.

3. Dataset Columns

Sepal length (cm): Length of the sepal, a flower part outside the petals.

Sepal width (cm): Width of the sepal.

Petal length (cm): Length of the petal, the usually colourful flower part.

Petal width (cm): Width of the petal.

flower_type: The target label identifying the flower species.

4. What the Uploaded Notebook Does

The uploaded deep-learning.ipynb notebook loads the built-in Iris dataset from scikit-learn and converts it into a pandas DataFrame.

It displays the dataset shape and column names, checks each column for missing values, and counts how many samples belong to each flower type.

Note: The current notebook is for dataset loading and exploration. It does not yet build or train an ANN model.

5. How to Run the Notebook in Kaggle

1. Open Kaggle and create or open a Notebook.

2. Add the code cells from deep-learning.ipynb, or upload the notebook if your Kaggle interface supports it.

3. Run each cell in order using Shift + Enter.

4. Review the table, dataset shape, column names, missing-value counts, and class counts.

6. Code Explanation

Cell 1 — Load the dataset and create a table:

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```
from sklearn.datasets import load_iris
import pandas as pd

iris = load_iris()

df = pd.DataFrame(
```

```
    iris.data,
    columns=iris.feature_names
)

df["flower_type"] = [
    iris.target_names[i] for i in iris.target
]

df.head()
```

Cell 2 — Check the dataset shape, columns, and missing values:

```
print("Rows and columns:", df.shape)
print("Column names:", df.columns.tolist())
print("Missing values:")
print(df.isnull().sum())
```

Cell 3 — Count samples in each class:

```
print(df["flower_type"].value_counts())
```

7. ANN Workflow (Next Step)

To train an ANN, the four measurement columns are used as input features (X), while flower_type is the target (y). The data should be split into training and testing sets. Feature scaling is commonly applied before training. An ANN can then be built with input, hidden, and output layers, trained on the training data, and evaluated on the test data.

For a three-class prediction task, the output layer commonly has three units with softmax activation. The model's predictions can be compared with the true labels using accuracy and a classification report.

8. Important Terms

Feature: An input measurement used by the model.

Target/label: The answer the model learns to predict.

Training data: Data used to teach the model.

Testing data: Separate data used to evaluate the trained model.

Classification: Predicting a category, such as a flower species.

Accuracy: The proportion of predictions that are correct.

9. Conclusion

The Iris dataset is a simple and useful starting point for machine learning. The uploaded notebook successfully focuses on loading and exploring the dataset. The next stage is to prepare the features and labels, build an ANN, train it, and evaluate its predictions.

Sample Dataset Preview

Sepal length	Sepal width	Petal length	Petal width	Flower type
5.1	3.5	1.4	0.2	setosa
6.4	3.2	4.5	1.5	versicolor
6.3	3.3	6.0	2.5	virginica